

Teacher Notes: Grading and Teaching Style

Probability and Statistics Foundations - Lectures 1-4

Teaching stance

Teach probability as a process of modelling uncertainty, not as a catalogue of formulas. Move repeatedly through four phases: preknowledge (what intuition suggests), formalization (events, variables, and assumptions), challenge (calculation or simulation), and verification (interpretation and reasonableness). Keep probability and statistics connected: assumptions generate distributions, while observations help evaluate those assumptions.

Preferred lesson pattern

1. Elicit a prediction. Ask students to estimate the direction or size of an answer before computing.
2. Name objects explicitly. Define the experiment, sample space, events, random variable, support, and parameters.
3. Show the representation. Use a tree, table, Venn diagram, PMF, density, or CDF when it clarifies structure.
4. Derive without silent steps. State the rule, substitute values, simplify, and interpret.
5. Verify. Check bounds, totals, units, edge cases, and whether the answer matches intuition.

Teaching style

Use direct explanations and frequent low-stakes questions. Ask students to explain why a method applies before asking for arithmetic. Treat incorrect answers as evidence about the current mental model. Contrast near-miss concepts explicitly, especially conditional direction, independence versus mutual exclusivity, density versus probability, and expected value versus a guaranteed outcome.

High-value teacher prompts

- “What is the conditioning event, and what becomes the new sample space?”
- “Which events partition all possible cases?”
- “Does order matter, and can an item be selected more than once?”
- “Is this value a probability, a density, or an outcome?”
- “What assumption makes the binomial model legitimate?”
- “What would change if neighboring observations were dependent?”

Common misconceptions and responses

Misconception	Instructional response
$P(A B)=P(B A)$	Use a two-way table or natural frequencies before introducing Bayes notation. Ask students to identify the denominator population in words.
Independent means mutually exclusive	Show that mutually exclusive positive-probability events cannot be independent because observing one rules out the other.
PDF height is a probability	Emphasize area over an interval. Use units: density has probability per unit, whereas area is dimensionless probability.
Expected value must be observable	Use repeated-trial averages and examples such as 0.5 children or 2.7 alarms to separate long-run mean from a single outcome.
A named distribution is automatically appropriate	Require a sentence matching each assumption to the context: fixed n , common p , independence, support, or constant density.

Grading principles

Grade the reasoning chain, not only the final number. Separate marks for setup, method, execution, interpretation, and verification. A correct answer with no visible reasoning should not automatically receive full credit when the assessment target includes modelling or explanation.

Recommended allocation within a multi-step item

Component	Typical share	Evidence
Definition/setup	20-30%	Correct events, variable, support, partition, or distribution parameters.
Method	25-35%	Correct rule or model and a coherent substitution.
Accuracy	20-30%	Algebra, arithmetic, and appropriate rounding.
Interpretation/check	15-25%	Contextual statement, assumption, limitation, or reasonableness check.

Partial credit and error propagation

Award method marks when the correct structure is visible, even after an arithmetic slip. Use follow-through for later parts that correctly use an earlier incorrect value. Do not award full credit for a formula that is merely written but not connected to the problem. Do not penalize a student twice for the same reversed conditional probability.

What earns interpretation marks

The student must identify what the numerical result refers to, including the conditioning population when relevant. “0.57” alone is not an interpretation. “Given an alarm, the model assigns about a 57% probability to high water” is. For expectation, require language such as long-run average or expected count, not a guaranteed prediction.

Feedback style

Feedback should be specific, actionable, and limited to the highest-leverage next step. Prefer “Your denominator should represent all alarm cases; compute $P(A)$ first” over “Review Bayes.” Prefer “Probability is area; multiply interval width by density height” over “Incorrect PDF.”

Academic integrity and AI-assisted work

When AI tools are permitted, assess the student's ability to verify and explain the result. Ask for event definitions, assumptions, intermediate equations, and a brief error check. Generic polished prose should not substitute for evidence of understanding. For take-home work, include a short oral check or personalized parameter variation when authenticity matters.

Lesson-by-lesson emphasis

Lecture 1 - Uncertainty. Prioritize translation between words, sets, and diagrams. Students should understand why probability answers are constrained and how complements or partitions simplify problems.

Lecture 2 - Conditional probability and Bayes. Use natural frequencies and tables before compact formulas. Make the direction of conditioning visible. Connect permutations and combinations to whether order matters.

Lecture 3 - Random variables and distributions. Emphasize that a random variable is a function assigning numbers to outcomes. Contrast PMFs, PDFs, and CDFs using the same example where possible.

Lecture 4 - Modelling. Require assumption statements. Use medians and quantiles through the CDF. Compare model fit and convenience, and discuss how dependence or non-constant probabilities affect conclusions.

Checks before releasing an assessment

1. Every question has a clear target and enough information.
2. Numerical values produce manageable calculations.
3. Marks align with the amount of reasoning required.
4. A complete mark scheme includes alternative valid methods and follow-through rules.
5. No item depends on ambiguous wording such as “accuracy” without defining whether it means conditional correctness.
6. The assessment samples calculation, interpretation, and modelling judgement.

Minimum standard for a passing response

A passing student should correctly identify the relevant probability structure in routine problems, carry out at least the central calculations, and give a minimally correct contextual interpretation. Persistent reversal of conditional probabilities, inability to distinguish density from probability, or selection of models without assumptions indicates that core outcomes have not yet been met.

Recommended closing routine

End each class with three questions: “What changed in your intuition?”, “Which definition or assumption did the most work?”, and “How could this model fail?” These prompts reinforce that formal computation and critical modelling belong together.